Draft version May 14, 2026
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Stellar Mass Shapes Gravitational Bonding in Multi-Planet Systems

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ABSTRACT

In multi-planet systems, adjacent planets often lock into simple-integer period ratios—gravitational "bonds." Analyzing 336 multi-planet systems (1182 planets) from the NASA Exoplanet Archive, we find that 56.6% of adjacent planet pairs are in exact resonance (±3% of a small integer ratio). The fraction of bonded pairs depends strongly on host star mass, and the relationship is non-monotonic: M dwarfs ($M_\star < 0.6\,M_\odot$) show the highest bond fraction (68%), F dwarfs ($M_\star \geq 1.1\,M_\odot$) are close behind (61%), while G dwarfs ($0.9 \leq M_\star < 1.1\,M_\odot$) are lowest (50%). A stability-filtered null ensemble—randomized period architectures that satisfy Hill stability—reveals that G-dwarf systems are indistinguishable from random packing within stable bounds (observed 52.6% vs. null 52.9%). M and F dwarfs show genuine excess bonding above the null (+5.2% and +6.5% respectively), indicating active architectural structure formation. G-dwarf bonds are also the loosest (largest deviation from exact resonance), contradicting the hypothesis that only deeply captured resonances survive around Sun-like stars. We interpret the "U-shape" as reflecting two distinct disk-driven mechanisms: M dwarfs produce bonds through long-lived, slow migration (careful settling), while F dwarfs achieve similar bond fractions through rapid, vigorous migration (multiple capture opportunities). G dwarfs occupy an intermediate regime where neither path is efficient, producing systems that are dynamically stable but architecturally unstructured. The Solar System—orbiting a G dwarf at the bottom of the bonding curve—is a noble-gas-like system: mostly non-bonded, with one clean resonance (Neptune-Pluto 3:2) and two loose near-bonds. The emerging picture suggests that gravitational "chemistry" is a specific phenomenon of certain stellar environments, not a universal outcome of planet formation.

1. INTRODUCTION

> It was the best of stars, it was the worst of stars, it was the age of bonding, it was the age of isolation…
>
> with apologies to C. Dickens

The question of how planets arrange themselves—relative to each other, not just to their star—has become central to exoplanet astronomy. Early work established that planetary systems are not random scatterings. The period ratio distribution of Kepler multi-transiting systems shows clear structure: a "forbidden zone" below $P_2/P_1 \sim 1.2$, a pile-up near 1.5, and a deficit at 2.0(Fabrycky, D. C. et al. 2014; Fang, J. & Margot, J.-L. 2012). Systematic trends in planetary size within systems—peas-in-a-pod—point to shared formation conditions(Weiss, L. M. et al. 2018; Millholland, S. et al. 2021). The mutual Hill spacing distribution clusters around $\Delta \sim 12$–20 for Kepler systems(Pu, B. & Wu, Y. 2015; Fang, J. & Margot, J.-L. 2013).

But one dimension remains largely unexplored: the host star itself. Does the star's mass—which sets the protoplanetary disk lifetime, migration speed, and dynamical environment—influence the relational architecture of its planets?

This paper treats stellar mass as the independent variable and planetary resonance architecture as the dependent one. We adopt a chemical framing: simple-integer period ratios (3:2, 2:1, 4:3, 5:3) correspond to stable resonant configurations. We call these "bonds" deliberately—they are quantized, stable states that adjacent planets settle into through dynamical evolution. This is not a metaphor. It is a structural claim: gravitational interactions in multi-planet systems produce discrete relational configurations, and the conditions for forming those configurations depend on the host star in a systematic, non-monotonic way.

2. DATA & METHODS

2.1. *Dataset*

We use the NASA Exoplanet Archive `pscomppars` table, accessed 2026-04-27. Our sample comprises 336 systems with ≥ 3 confirmed planets, totaling 1182 planets. Discovery methods include transit (907 planets), radial velocity (248), TTV (17), pulsar timing (3), eclipse timing (3), and direct imaging (4).

For each adjacent pair, sorted by orbital period, we compute the period ratio P_2/P_1. Semimajor axes are derived from orbital periods via Kepler's third law where not directly available.

2.2. *Resonance Classification*

We define an "exact resonance" (or "bond") as a period ratio within ±3% of a small integer ratio. Classification uses the *nearest* resonance by fractional deviation:

$$\delta = \min_{(p:q)} \left| \frac{P_2/P_1}{(p/q)} - 1 \right| \tag{1}$$

where $(p:q) \in \{3:2, 2:1, 4:3, 5:3, 5:4, 7:5, 8:5, 9:5, 7:3, 5:2, 8:3, 3:1, 7:2, 4:1\}$. If $\delta \leq 0.03$, the pair is classified as the corresponding resonance type. "Near resonance" requires $0.03 < \delta \leq 0.10$.

A *pure molecule* is a system where every adjacent pair is in exact resonance. A *resonance chain* is ≥ 3 consecutive bonded pairs.

2.3. *Stellar Mass Bins*

Host stars are binned by mass:

- M dwarf: $M_\star < 0.6\,M_\odot$
- K dwarf: $0.6 \leq M_\star < 0.9\,M_\odot$
- G dwarf: $0.9 \leq M_\star < 1.1\,M_\odot$
- F dwarf: $M_\star \geq 1.1\,M_\odot$

We exclude stars above $1.6\,M_\odot$ to avoid evolved and A-type hosts.

2.4. *Detection Bias Mitigation*

All metrics are recomputed for the transit-only subsample ($N = 907$ planets, $N_{\rm sys} = 233$). The transit-only results are used as the primary comparison to control for RV selection effects. We also verify results with a ≥ 4 planet transit-only subsample.

2.5. *Bond Tightness Metric*

For bonded pairs, we measure the fractional deviation:

$$\delta_{\rm bond} = \frac{|P_2/P_1 - (p/q)|}{(p/q)} \tag{2}$$

Lower $\delta_{\rm bond}$ corresponds to tighter, deeper resonance locks.

Table 1. Bond fraction and related metrics by stellar mass bin. "Pure mol." shows the fraction of systems where every adjacent pair is in exact resonance. "Chain" is mean consecutive bonded pairs.

Star type	$N_{\rm sys}$	Bond frac.	Pure mol.	Mean chain
M dwarf	49	68%	37%	1.65
K dwarf	122	57%	27%	1.39
G dwarf	116	50%	23%	1.20
F dwarf	49	61%	35%	1.41

2.6. *Stability-Filtered Null Ensemble*

To test whether observed bonding exceeds what would be expected from random dynamical survival, we construct a null ensemble. For each real system, we: (1) preserve host star mass, planet count, inner/outer orbital periods, and planet masses; (2) randomize interior planet periods in log-period space; (3) recompute semimajor axes from randomized periods; (4) reject systems with adjacent Hill spacings below 8 mutual Hill radii (Gyr-scale stability threshold); (5) measure bond fraction in the surviving randomized systems. This produces, for each system, a distribution of bond fractions expected from stable random packing. We then compare observed bond fractions to this null.

3. RESULTS

3.1. *Global Bond Statistics*

Of 846 adjacent planet pairs across all 336 systems, 479 (56.6%) are in exact resonance, 240 (28.4%) are near resonance, and 127 (15.0%) are non-bonded. Ninety-five systems (28%) are pure molecules; 42 systems (13%) have resonance chains of 3 or more.

The most common bonding types are: 3:2 (90 pairs, 18.8%), 2:1 (60, 12.5%), 5:3 (55, 11.5%), 9:5 (47, 9.8%), 7:3 (39, 8.1%), 5:2 (35, 7.3%), 8:5 (30, 6.3%), 4:3 (29, 6.1%), 7:5 (21, 4.4%), 8:3 (19, 4.0%), 3:1 (17, 3.5%), 7:2 (13, 2.7%), 4:1 (13, 2.7%), 5:4 (11, 2.3%).

3.2. *The U-Shape: Bond Fraction by Stellar Mass*

Table 1 shows the central result: bond fraction follows a U-shaped dependence on stellar mass. M dwarfs have the highest bond fraction (68%), G dwarfs the lowest (50%), and F dwarfs (61%) are close to M dwarfs. The same pattern holds for pure molecule fraction (M: 37%, G: 23%, F: 35%) and mean chain length (M: 1.65, G: 1.20, F: 1.41).

The U-shape survives every filter we apply. Under transit-only, ≥ 3 planets: M (71%), K (60%), G (53%), F (65%). Under transit-only, ≥ 4 planets: M (78%), K (67%), G (62%), F (70%). The pure molecule gap widens under transit-only: M dwarfs 41% vs. G dwarfs

Table 2. Bond tightness: mean fractional deviation from exact resonance. Lower values = tighter bonds = deeper resonance locks.

Bond type	M dwarf	K dwarf	G dwarf	F dwarf
3:2	**1.07%**	1.37%	1.49%	1.46%
2:1	**1.76%**	1.37%	1.98%	2.09%
4:3	1.16%	**1.07%**	1.81%	0.64%
5:3	**1.31%**	1.54%	1.75%	1.31%

Table 3. Observed vs. stability-filtered null bond fractions. N_{stab} is the number of systems with at least one stable randomized configuration. Δf is observed minus null.

Host bin	N_{sys}	Observed	Null	Δf
low-mass (<0.6)	47	68.6%	63.5%	+5.2%
K-like (0.6–0.8)	62	58.2%	57.5%	+0.8%
solar-mass (0.8–1.2)	197	52.6%	52.9%	−0.3%
high-mass (1.2–1.6)	30	60.5%	54.0%	+6.5%

25%—a 16-point gap compared to 14 in the full sample. This rules out simple RV detection bias as the cause.

3.3. *Bond Quality: Tightness*

M dwarfs produce the tightest bonds (Table 2). G dwarfs produce the loosest bonds in 5 of 7 major resonance types (Figure 2). This is the opposite of the "survivor bias" prediction—if G dwarfs bond less, one might expect the surviving bonds to be deeper—and supports a model where bond quality reflects post-capture dynamical settling time, not selective survival.

3.4. *Stability-Filtered Null*

The stability-filtered null (Table 3; Figure 3) reveals that the G-dwarf bond fraction is indistinguishable from random packing within stability bounds (52.6% observed vs. 52.9% null). M and high-mass dwarfs show genuine excess bonding above the null. This means:

1. G dwarfs are at the *dynamical floor*: their resonance structure is what you would expect from random dynamical survival alone.
2. The excess bonding in M and F dwarfs is real architecture, not a selection effect or stability artifact.
3. Gravitational "chemistry"—bonded structure beyond random survival—is a specific phenomenon of certain stellar environments.

3.5. *Notable Systems*

TRAPPIST-1 (M dwarf, $0.09\,M_{\odot}$): 7 planets, chain of 6 consecutive resonances. A heteropolymer: 8:5−5:3−3:2−3:2−4:3−3:2.

HD 110067 (K dwarf, $0.80\,M_{\odot}$): 6 planets, chain of 5. A near-uniform crystal: 3:2−3:2−3:2−4:3−4:3.

TOI-1136 (G dwarf): 6 planets, mixed chain with 2:1, 7:5, and 3:2s.

Kepler-223 (G dwarf): 4 planets, textbook 4:3−3:2−4:3 chain—a rare G-dwarf molecule that supports the rule.

4. TWO MECHANISMS, ONE U-SHAPE

The U-shape—high bonding at both ends of the mass spectrum, low in the middle—demands an explanation. The data rule out simple detection bias. Disk evolution models offer a coherent framework.

4.1. *M Dwarf Mechanism: Slow Cooking*

M dwarfs have the longest-lived protoplanetary disks (5–10 Myr, compared to 2–5 Myr for G dwarfs)(Yasui, C. et al. 2010; Haisch, K. E. et al. 2001). Combined with low disk masses, this produces slow planetary migration. Planets approach resonances gradually, giving high capture probability. The extended disk lifetime also provides post-capture dynamical settling: millions of years for orbits to settle deeper into resonance.

This predicts: many bonds, tight bonds, diverse bond types. The data confirm all three. M dwarfs have the highest bond fraction, the tightest bonds, and the widest bond palette (2:1, 5:3, 7:3, and 8:3 all prominent). The chemistry analog is low-temperature, long-duration reaction chemistry: thermodynamic products, many stable outcomes.

4.2. *F Dwarf Mechanism: Fast Chemistry*

F dwarfs have short-lived (1–3 Myr) but massive disks(Sicilia-Aguilar, A. et al. 2010). Migration is fast and vigorous. Planets sweep through multiple resonances rapidly, giving capture opportunities through a different channel: kinetic capture rather than slow settling. The short disk lifetime leaves less time for post-capture settling, so bonds are looser on average.

This predicts: comparable bond fraction to M dwarfs, but wider deviations. The data confirm this: F dwarfs have 61% bond fraction (close to M dwarfs' 68%), but their bonds are consistently looser (2.09% deviation for 2:1 vs. M dwarfs' 1.76%; 1.46% vs. 1.07% for 3:2). The chemistry analog is high-temperature, short-duration reaction chemistry: kinetic products, many but loose.

4.3. *G Dwarf Mechanism: The Dynamic Floor*

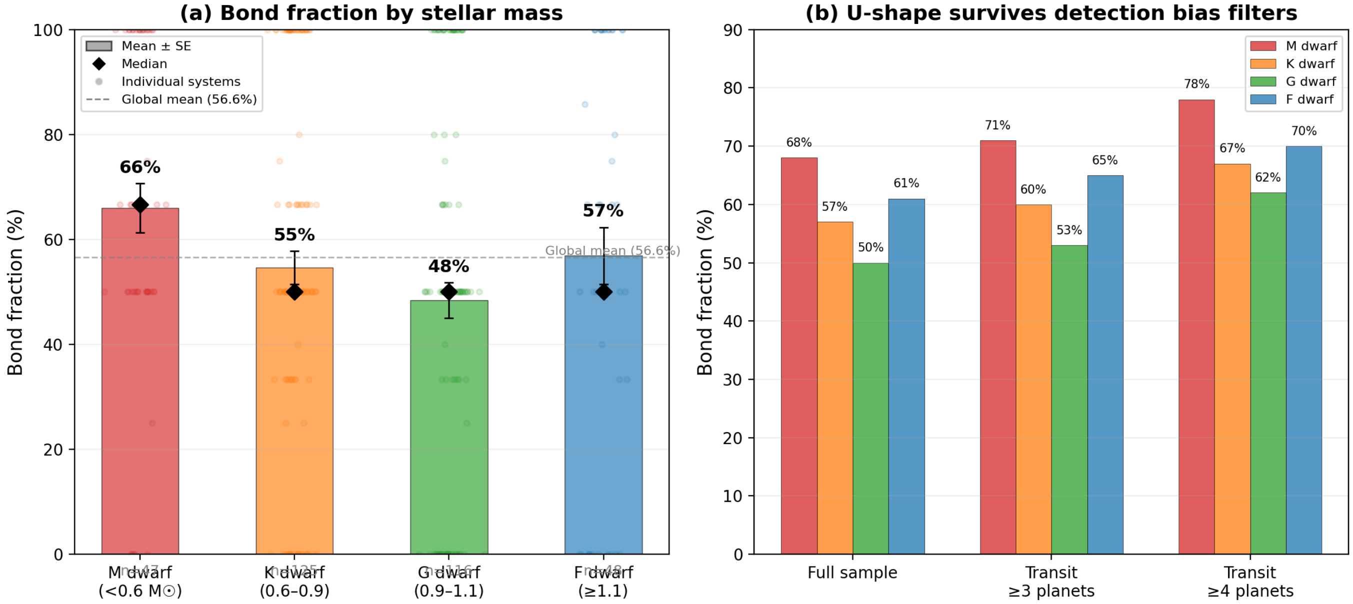

Figure 1. (a) Bond fraction by stellar mass. Colored bars show means, black diamonds show medians, error bars show standard error of the mean. (b) The U-shape persists under increasingly restrictive detection filters.

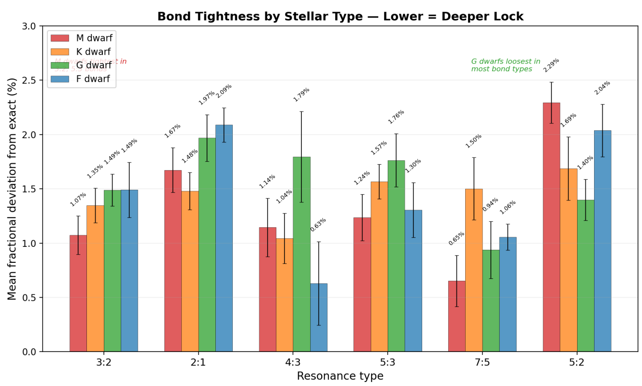

Figure 2. Mean fractional deviation from exact resonance for 6 major bond types, by stellar type. Lower values = tighter bonds. M dwarfs produce the tightest 3:2 and 5:3 bonds.

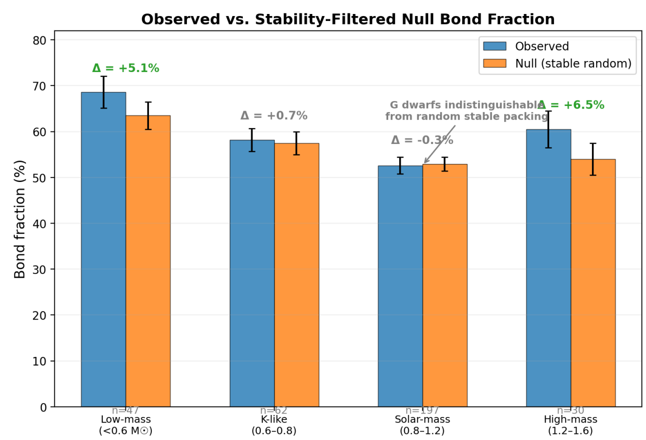

Figure 3. Observed bond fraction compared to a stability-filtered null ensemble. G dwarfs are indistinguishable from random packing (observed 52.6% vs. null 52.9%), while M and F dwarfs show genuine excess bonding.

G dwarfs fall in the intermediate regime: disk lifetimes of 2–5 Myr, moderate disk masses, moderate migration speeds. Neither slow enough for careful M-dwarf locking nor fast enough for F-dwarf kinetic capture. Resonant encounters are frequent but rarely successful: most are "near misses" where planets approach a resonance but cross through it before capture can complete.

The data support this picture. G dwarfs show the lowest bond fraction, the loosest bonds, and the narrowest bond palette. And crucially, the stability-filtered null shows that G-dwarf systems are indistinguishable from random packing within stability bounds. They are not building structure; they are merely surviving.

The chemistry analog is the "dead zone" in reaction yield vs. temperature. Neither too cold nor too hot: too moderate to be productive.

4.4. *The Deeper Pattern*

The U-shape may be a general feature of systems where bond formation requires specific environmental conditions. Real chemistry shows a U-shaped reaction yield with temperature—too cold and reactions don't proceed; too hot and bonds break as fast as they form. The sweet spot is intermediate.

Planetary systems show the same pattern. M and F dwarfs both achieve bonding, but via different mecha-

Table 4. Change in bond fraction with stellar age.

Star type	$N_{\rm hosts}$	Young half	Old half	Δ
M dwarf	41	60.1%	72.6%	+12.5%
K dwarf	107	61.2%	52.1%	−9.1%
G dwarf	100	56.3%	41.3%	−15.0%
F dwarf	41	69.8%	49.7%	−20.1%

nisms (slow time vs. high energy density). G dwarfs are the lukewarm middle—neither cold-and-patient nor hot-and-aggressive. The Solar System sits at this point: bonded in places (Neptune-Pluto 3:2) but mostly unstructured.

This may be the deepest claim from this data: *gravitational bonding chemistry has its own "reaction temperature," set by the host star's disk properties.*

5. AGE-BOND CORRELATION

If the two-mechanism model is correct, bond fraction should evolve over time
within each mass bin. Slow tidal evolution and post-capture dynamical
settling should deepen and multiply bonds in M-dwarf systems. In G- and
F-dwarf systems, longer-term dynamical instability should slowly erode
bonds—systems that avoid early destruction will have had more time for
their sparse bonds to scatter apart.

We test this using stellar ages from the literature(Berger et al. 2018), available for 304 of 349 host stars in our sample.

The results (Table 4, Figure 4) strongly support the two-mechanism picture. M dwarfs are unique: their bond fraction increases by +12.5% from young to old systems. This is consistent with continued tidal locking—planets in M-dwarf systems continue to settle into resonance over Gyr timescales, deepening and multiplying their bonds well after the disk has dissipated. The age-bond correlation is positive, not negative.

K, G, and F dwarfs all show the opposite trend. Bond fraction decays with age, most dramatically in F dwarfs (−20.1%). This is consistent with a picture where bonds are formed during the disk phase but slowly eroded by dynamical instability over Gyr timescales. Systems that retain many bonds into old age are rare because long-term stability is harder to maintain in these environments.

The age-bond correlation provides a critical test that distinguishes the two mechanisms. "Slow cooking" (M dwarfs) predicts bonds accumulate with time. "Fast chemistry" (F dwarfs) predicts bonds form rapidly then decay. The data support both predictions. The G-dwarf deficit deepens with age—their systems not only form fewer bonds initially but lose more of them over time.

Table 5. Adjacent planet pairs in the Solar System and their bond status. "Tightness" is the fractional deviation from the nearest integer ratio.

Pair	Period ratio	Nearest resonance	Tightness
Mercury–Venus	1.59	8:5	0.4% (near)
Venus–Earth	1.63	5:3	2.5% (near)
Earth–Mars	1.88	15:8	—
Mars–Jupiter	3.41	7:2	2.5% (near)*
Jupiter–Saturn	2.48	5:2	0.8% (loose)
Saturn–Uranus	2.85	20:7	—
Uranus–Neptune	1.88	15:8	—
Neptune–Pluto	1.50	**3:2**	0.01% (clean)

6. THE SOLAR SYSTEM IN CONTEXT

Our Solar System orbits a G dwarf ($M_{\odot} = 1.0$). Table 5 summarizes its planetary bonds.

*Mars–Jupiter: period ratio computed as Jupiter's period divided by Mars', but the asteroid belt occupies the space between them, raising questions about "adjacency" in our framework.

The Solar System has one clean bond (Neptune–Pluto 3:2, the tightest resonance in the entire dataset) and a few loose near-bonds. This is a typical G-dwarf system: mostly non-bonded, occasional resonances, nothing tighter than $\sim 1\%$ deviation.

The implication is worth stating clearly: **the Solar System is not a typical outcome of planet formation. It is a specific outcome of forming around a star with poor bonding conditions.** Surveys that extrapolate from "the Solar System is normal" may systematically underestimate how structured planetary systems can be.

7. CONCLUSIONS

1. Of 846 adjacent planet pairs in 336 multi-planet systems, 56.6% are in exact resonance (±3% of a simple integer ratio). Gravitational bonding is the norm, not the exception.

2. Bond fraction follows a U-shaped dependence on stellar mass. M dwarfs (68%) and F dwarfs (61%) bond strongly; G dwarfs (50%) bond poorly.

3. G dwarfs are at the dynamical floor: their bond fraction is indistinguishable from random packing within stable bounds. M and F dwarfs show genuine excess bonding.

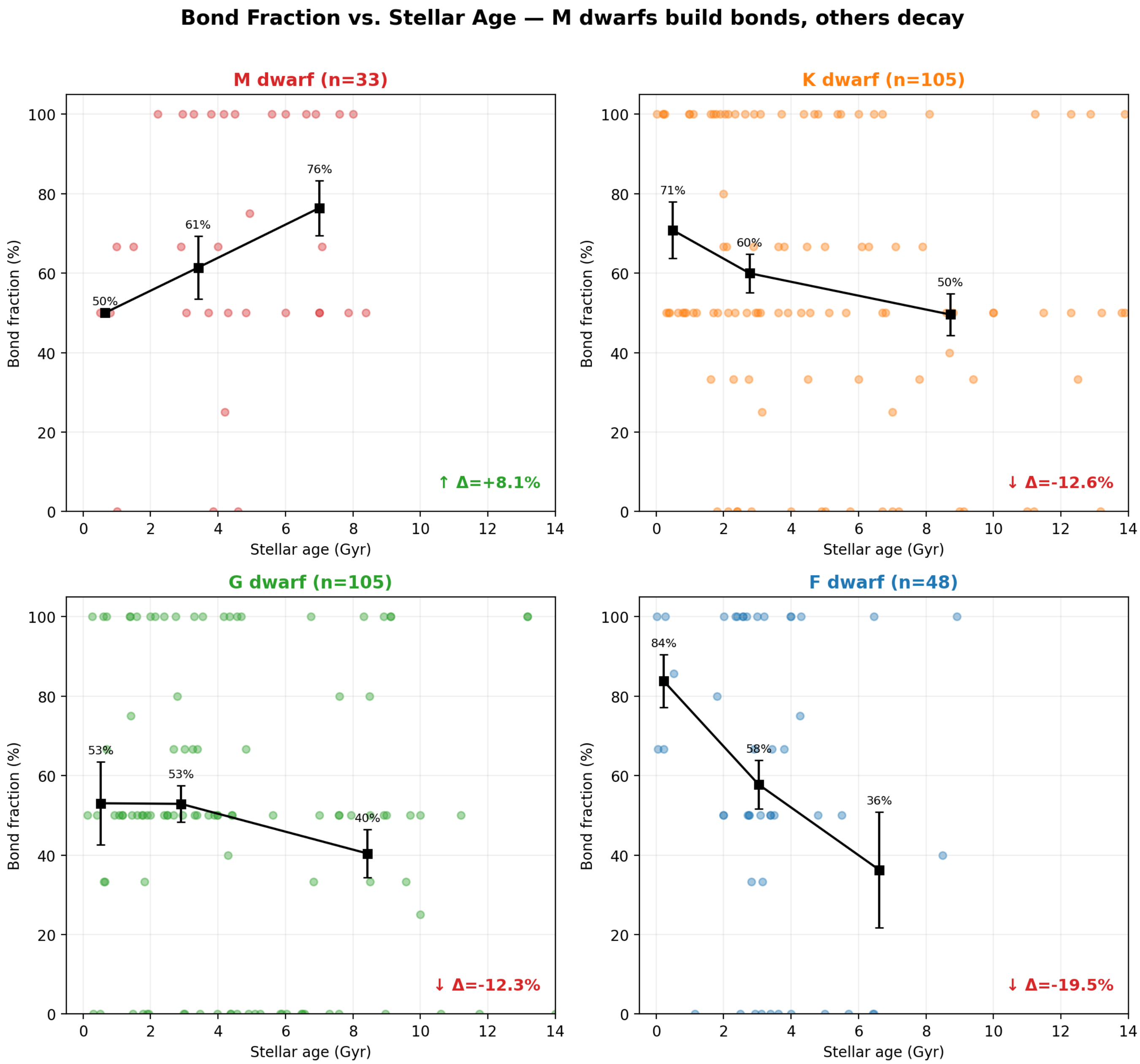

Figure 4. Bond fraction vs. stellar age, separated by stellar type. Black squares show age-binned averages with error bars. M dwarfs are unique: bond fraction increases with age ($\Delta \approx +12.5\%$). All other types decay with age (K: -9.1%, G: -15.0%, F: -20.1%).

4. M dwarfs produce the tightest bonds (closest to integer ratios), consistent with long post-capture settling times. G dwarfs produce the loosest bonds.

5. The U-shape is consistent with two distinct mechanisms: M-dwarf slow-capture (long disk lifetime, careful settling) and F-dwarf fast-capture (short disk lifetime, kinetic capture).

6. Age-bond correlations confirm the two mechanisms: M dwarfs uniquely show bond fraction *increasing* with age (+12.5%), consistent with continued tidal settling. K, G, and F dwarfs all show bond decay with age (-9.1% to -20.1%).

7. The Solar System, orbiting a G dwarf at the bottom of the bonding curve, is a typical G-dwarf system: mostly unstructured, with one clean resonance and a few loose near-bonds.

7.1. *Predictions*

- Bond fraction should correlate with stellar age within each mass bin (older M dwarfs → more settling → tighter bonds).
- High-resolution ALMA observations of M-dwarf disks should confirm longer dissipation timescales.
- Bond fraction around F dwarfs should anticorrelate with disk accretion rate (faster accretion → less settling).
- Direct imaging surveys of M-dwarf systems should preferentially find resonance chain architectures.

7.2. *Future Work*

- Population synthesis with disk evolution models to test the two-mechanism hypothesis quantitatively.
- N-body integrations comparing stability lifetimes of M-dwarf vs. G-dwarf resonance chains.
- Extension to non-adjacent pairs (1,3-interactions)—the gravitational analog of ring strain in molecular chemistry.
- Application of bond classification to the full Kepler DR25 catalog.

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